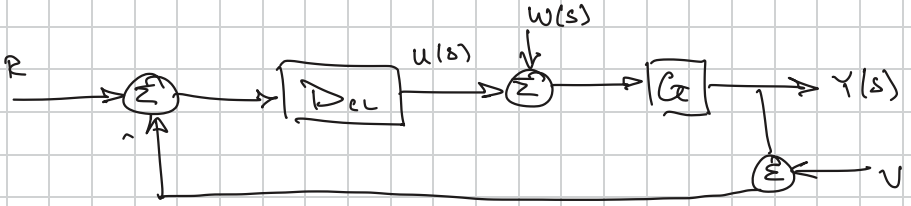
Lecture #11 / 8 October 2026

HW #2 due next Thursday

NO CLASS next Tuesday

Canonical Feedback Loop



R = reference
Y = system state
} Want reference and the system state to converge (tracking)

W = disturbance input
} Want the system state to reject the disturbance (regulating)

V = measurement noise
} We want the system state to filter the measurement noise (filtering)

The key takeaway: there is no free lunch: you cannot achieve all three of these perfectly at the same time

Tuesday, we learned:

sensitivity function
$$S = \frac{1}{1+GD_{cl}} = \frac{E}{R}$$

complementary sensitivity function
$$T = \frac{GD_{cl}}{1+GD_{cl}} = \frac{Y}{R} = \frac{Y}{V}$$

load sensitivity
$$GS = \frac{G}{1+GD_{cl}} = \frac{Y}{W}$$

noise sensitivity
$$D_{cl}S = \frac{D_{cl}}{1+GD_{cl}} = \frac{U}{R} = \frac{U}{V}$$

$$\boxed{T+S=1}$$

why there is no free lunch!

- Open loop control cannot modify pole locations
  ⇒ Wherever the plant poles lie are what you're stuck with
- Closed loop control allows us to place the loop's poles
  → There are special conditions on *where* we can place the poles.

Example (revisited)

Tracking: $\frac{Y}{R} \to 1$

(when $\frac{Y}{R} \to 1$, then $R \to Y$)

$$G(s) = \frac{1}{s^2+3s+9} \qquad D_{CL} = \frac{C_2 s^2 + C_1 s + C_0}{s(s+d_1)}$$

↑ the closed-loop system

We want char. poly. to be $\underbrace{(s+6)}_{s=-6}\underbrace{(s+3)}_{s=-3}\underbrace{(s^2+3s+9)}_{=0} = \emptyset$

Want to solve for C_0, C_1, C_2, d_1 such that this is the closed char. poly.

$$T = \frac{Y}{R} = \frac{GD_{CL}}{\underbrace{1+GD_{CL}}_{=0}} \Rightarrow 1 + GD_{CL} = \emptyset$$

$$\Rightarrow 1 + \left(\frac{1}{s^2+3s+9}\right)\left(\frac{C_2 s^2 + C_1 s + C_0}{s(s+d_1)}\right) = \emptyset$$

$$\Rightarrow \underbrace{s(s^2+3s+9)(s+d_1) + C_2 s^2 + C_1 s + C_0 = \emptyset}$$

Expand it $a_4 s^4 + a_3 s^3 + a_2 s^2 + a_1 s + a_0 = \underbrace{(s+6)(s+3)(s^2+3s+9)}$

expand

$s^4 + s^3$

$C_2 = 18$
$C_1 = 54$
$C_0 = 162$
$d_1 = 9$

$$\Rightarrow \quad D_{CL} = \frac{18(s^2+3s+9)}{s(s+9)} \qquad G = \frac{1}{s^2+3s+9}$$

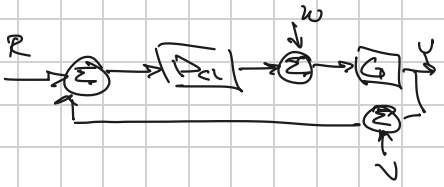
(for c_0, c_1, c_2, ϕ_1 plugged in)

POLES CANCEL

$$\rightarrow \quad 1 + GD_{CL} = \emptyset \Rightarrow 1 + \left[\frac{18(s^2+3s+9)}{s(s+9)}\right]\left[\frac{1}{s^2+3s+9}\right] = 0$$

$$\Rightarrow \quad s(s+9) + 18 = \emptyset$$

For tracking: $T = \frac{Y}{R} = \frac{GD_{CL}}{1+GD_{CL}} = \frac{18}{(s+3)(s+6)}$ (Poles Cancelled)



$S = \frac{E}{R} = \frac{1}{1+GD_{CL}} = \frac{s(s+9)}{(s+3)(s+6)}$ (Poles Cancelled)

(noise sensitivity) $D_{CL}S = \frac{U}{V} = \frac{D_{CL}}{1+GD_{CL}} = \frac{18(s^2+3s+9)}{(s+3)(s+6)}$ (Poles Cancelled)

(load sensitivity) $GS = \frac{Y}{W} = \frac{G}{1+GD_{CL}} = \frac{s(s+9)}{(s+3)(s+6)(s^2+3s+9)}$ (POLES NOT CANCELLED)

You cannot just calculate one transfer function and assume that all others will behave the same.

The disturbance response has an extra mode that isn't accounted for by the other transfer functions.

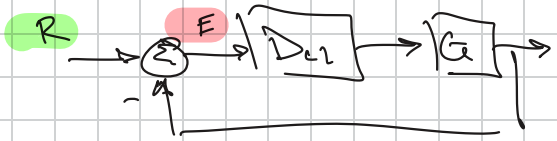
What about steady state error?

Still looking at tracking, but we're going to look at Sensitivity function to assess steady state tracking error.

$$S = \frac{s(s+9)}{(s+3)(s+6)} = \frac{E}{R}$$

We want to find what E does as $t \to \infty$ for various input references.

① Impulse:



$R = \delta(t)$

$R(s) = \mathcal{L}\{\delta(t)\} = 1$

$E = S \cdot R$ ($R = 1$)

$$\lim_{t\to\infty} e(t) = \lim_{s\to 0} sE(s) \quad \text{[Final Value Theorem]}$$

$$= \lim_{s\to 0} s \cdot S = \lim_{s\to 0} s \cdot \frac{s(s+9)}{(s+3)(s+6)} = 0$$

$\Rightarrow$ The error to an impulse on the reference will lead to zero steady state error.

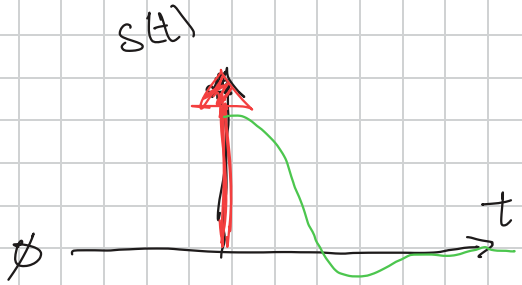
② Step: $r(t) = 1(t) \Rightarrow R(s) = 1/s$

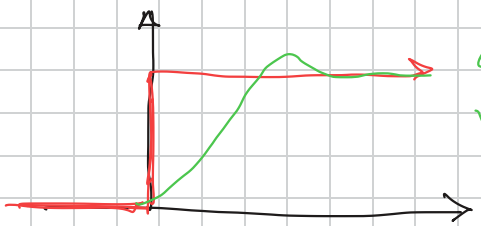
$$E = S \cdot R = \left(\frac{s(s+9)}{(s+3)(s+6)}\right) \cdot \frac{1}{s}$$

$$\lim_{t\to\infty} e(t) = \lim_{s\to 0} s \cdot S \cdot R = s \cdot \left(\frac{1}{s}\right)\left(\frac{s(s+9)}{(s+3)(s+6)}\right)$$

$$= 0$$

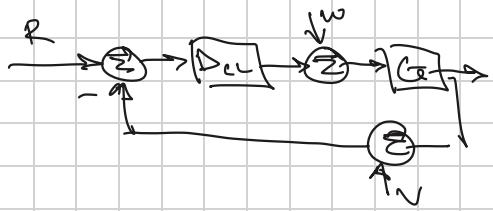
This system has zero steady state error to a step input as a reference.





error is zero $\Rightarrow$ reference and state converge as $t \to \infty$

$$\frac{Y}{W} = GS$$ (load sensitivity)



We want to show: if W is a constant disturbance AND if D_{CL} has a pole at the origin then the error to this disturbance is zero.

However: if G has a pole at the origin instead, the error due to this disturbance will not be zero.

$$D_{CL} = \frac{N(s)}{s \cdot D(s)} \quad \text{vs} \quad G = \frac{N(s)}{s \cdot D(s)}$$

These are not equivalent integrators!

You see the difference in the disturbance response.

Assume that $GD_{CL} = \frac{b(s)c(s)}{a(s)d(s)}$

$= \frac{b(s)}{a(s)}$ $= \frac{c(s)}{d(s)}$

this is why we add integrators in PI control

$$\Rightarrow G \cdot S = \frac{bd}{ad + bc}$$

bias term

$$\Rightarrow e_{ss} = GS(0) \cdot w_0 = w_0 \frac{b(0)d(0)}{a(0)d(0) + b(0)c(0)}$$

Integrator in the controller $\Rightarrow d(0) = 0$

$\Rightarrow e_{ss} = 0$

$e_{ss} \neq 0$

Integrator in the plant $\Rightarrow a(0) = 0 \Rightarrow \frac{b(0)d(0)}{b(0)c(0)} = \frac{d(0)}{c(0)}$

PID Control

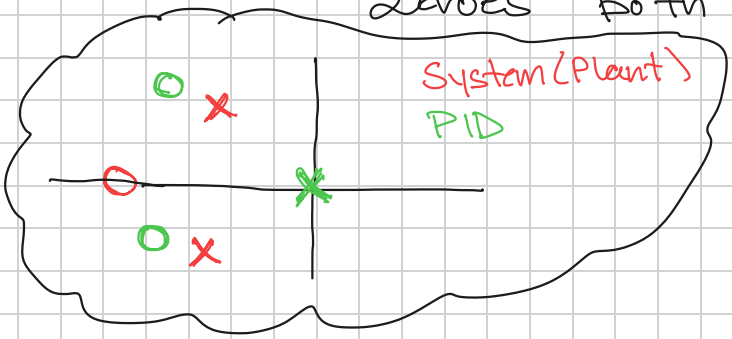


$$u(t) = K_p e(t) + K_I \int_0^t e(\tau)\,d\tau + K_D \dot{e}(t)$$

$$U(s) = \left[K_p + K_I/s + K_D s\right] E(s)$$

$$= \left[\frac{K_D s^2 + K_p s + K_I}{s}\right] E(s)$$

A PID controller adds a pole at the origin and two zeroes both in the open loop transfer function



open loop transfer function

Start with proportional control

$$u(t) = K_p e(t)$$

Assume that $G(s) = \frac{A}{s^2 + a_1 s + a_2} = \frac{A}{s^2 + 2\zeta\omega_n s + \omega_n^2}$ ← 2nd order plant

$$S = \frac{1}{1 + K_p G} \Rightarrow 1 + K_p\left(\frac{A}{s^2 + a_1 s + a_2}\right) = 0$$

← 2nd order closed loop system

$$\Rightarrow \boxed{s^2 + a_1 s + a_2 + A K_p = 0}$$

a_1 stays the same

added a term onto a_2

$$s^2 + a_1 s + a_2 + K_p A = 0$$

$$\boxed{s^2 + 2\zeta\omega_n s + \omega_n^2 = 0}$$ 2nd Order Form

the gist: K_p changed both the natural frequency and the damping

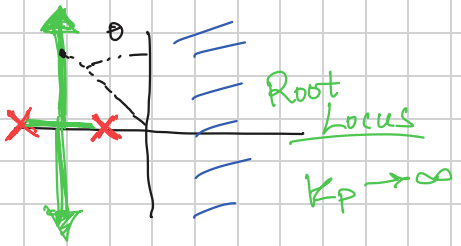
$$\omega_n^2 = K_p A + a_2$$

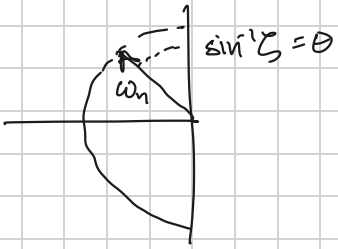
$$2\zeta\omega_n = a_1 \Rightarrow \zeta = \frac{a_1}{2\omega_n} = \frac{a_1}{2\sqrt{K_p A + a_2}}$$

$K_p \to \infty \Rightarrow \zeta \to 0$

(Assume $A > 0$) When $K_p \uparrow \infty \Rightarrow \omega_n \uparrow \infty$

$\Rightarrow \zeta \downarrow 0$





This implies that P control won't (necessarily) lead to any poles going to right half plane.

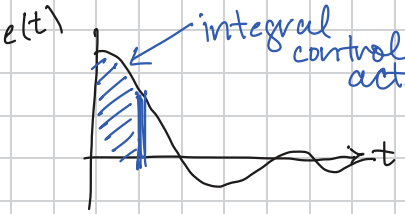
FLAW: there's rarely ever just two poles

Integral Control

P control vs just calculating the instantaneous error

I control is calculating the accumulated error

(they have memory and they remember the past)



Derivative Control $u(t) = K_D \dot{e}(t)$

(rate feedback)

- Pretty much never used alone.

→ PD control

$$K_P e(t) + K_D \dot{e}(t) = K_P \left(e(t) + T_D \dot{e}(t) \right)$$

$$\approx K_P e(t + T_D)$$

"Predicting" the future